

OH FLAME ABSORPTION/EMISSION SPECTROSCOPY

BY

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FINAL PROJECT REPORT

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ABSTRACT

This project focuses on using optical spectroscopy to understand the combustion behavior of several everyday fuels, specifically butane, candle wax, sterno gel, hexamine tablets, and propane, and to develop a physics-informed spectroscopic framework for characterizing these combustion environments using both UV broadband absorption spectroscopy (BAS) and chemiluminescence emission measurements. A broadband UV source and a spectrometer calibrated using a Hg lamp were used to capture four canonical spectra – lamp-on/flame-off, lamp-on/flame-on, lamp-off/flame-on, and lamp-off/flame-off to isolate both molecular absorption and spontaneous emission from OH radicals in the flame. The temperature and OH mole fraction distributions were obtained at different heights above the source. Robust preprocessing pipelines were implemented, including rolling-percentile masking and polynomial baseline removal, to eliminate continuum absorption and detector drift in absorption spectra, and to recover the chemiluminescence signals in emission spectra. Temperature determination was performed using a Boltzmann analysis of line-resolved absorbance and emission features, supported by parameters from the HITRAN and LIFBASE spectroscopic databases. To extend the method beyond a single effective temperature, a two-zone reconstruction model was designed to estimate spatial temperature distribution along the line of sight. The resulting workflow provides a low-cost approach for quantitative OH thermometry and flame structure estimation using compact optical hardware and data-driven spectral processing.

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CHAPTER 1: INTRODUCTION

The hydroxyl radical (OH) has been characterized as “God’s gift to the spectroscopist.” It shows up in many different systems, is generally free of interference, has great line spacing for temperature measurement, is not too far into the UV, and has large transition strengths. It’s also easily seen in emission and absorption. As such, it’s perhaps the most widely studied diatomic species in reacting systems.

CHAPTER 2: LITERATURE REVIEW

Luck et al. [1] conducted a comprehensive set of measurements within a single combustion zone: the actual temperature of the gas, the precise concentration of the OH radical, and the direct determination of the collision broadening parameter. Using a tunable, narrow-bandwidth dye laser, which was frequency-doubled using a KDP crystal, they targeted the UV spectral signature in the 1-0 band of the ($X^2\Pi-A^2\Sigma^+$) system of OH, located near 2830 Å, inside a laminar, flat methane-air flame. The laminar flow was confirmed via Schlieren imaging. From the recorded spectrum, two key parameters were derived: the concentration of the OH radical from the area under the Voigt profile, and the translational temperature from the full width at half maximum (FWHM) of the curve. The rotational temperature, calculated from the ratio of the two local line-absorption coefficients, was used to validate the existence of local thermal equilibrium (LTE) by comparison with translational temperature. The expected flat temperature profile, characteristic of flat flames, was confirmed experimentally. At the flame axis ($s = 20\text{mm}$), the temperature deviated from the adiabatic temperature by approximately 100K and decreased radially to 2050K at $s = 40\text{mm}$, which was attributed to the concentric airflow, used to stabilize the flame. The good agreement observed between the translational temperatures determined for the homogeneous and inhomogeneous systems justified the assumption that the flame is homogeneous. The OH-concentration was found to decrease on the flame axis from 2350 ppm at a height of $s = 20\text{ mm}$ to 1500 ppm at a height of $s = 40\text{ mm}$, a value five times higher than predicted under LTE at 2050 K. Finally, the collision broadening parameter, originally believed to be dependent only on the local OH concentration and the ground state rotational quantum number, was shown to also depend on the vibrational quantum number, as lower values were observed in the 1-0 band compared to the 0-0 band.

Yang et al. [2] outlined several advantages of UVBAS over TDLAS, including significantly higher accuracy, non-intrusive operation, precise OH measurements, higher sensitivity, and a very low detection limit for OH radicals. TDLAS suffers from a lower signal-to-noise ratio (SNR), high spectral interference from water molecules that absorb light almost across the entire IR spectrum, and ambient absorption from water and CO₂ molecules present in the ambient air outside the flame between the laser and the detector. UVBAS focuses on electronic transitions of diatomic radicals instead of the vibrational transitions used in IR, which are much stronger absorbers of light by around 2 orders of magnitude. Furthermore, spectral interference is also reduced in the UV region due to interfering molecules primarily absorbing in the IR region. BAS employs a broadband light source that covers a wide range of wavelengths simultaneously and analyses the absorption across the whole range, unlike TDLAS, which uses one or two very specific, narrow laser lines. This approach enhances the temperature sensitivity of the measurement. However, the study noted that UVBAS measurements are affected by spectrometer instrument broadening: as broadening increases, absorption decreases, SNR decreases, and the number of resolvable peaks decreases, which in turn lowers temperature sensitivity. Therefore, a high-resolution spectrometer is essential for accurate OH BAS measurements. The study examined 209 different OH absorption lines in the 0-0 band between 306 and 318 nm all at once. The absence of OH radicals in the ambient air further mitigates interference. Using a standard McKenna burner to create a flat premixed flame, the researchers tested a range of equivalence ratios from lean to rich. Results were benchmarked against two references: (1) 2D axisymmetric CFD simulations using the AramcoMech 1.3 chemical model and (2) direct comparison with TDLAS measurements based on a known CO absorption line pair. For stable flames, RMS error between observed temperature and CFD predictions was under 16K, and OH concentration error was below 150 ppm.

The average error observed between UVBAS and TDLAS measurements was about 31.5 K. Notably, TDLAS failed at lean equivalence ratios due to reduced concentration of CO, whereas UVBAS performed consistently across all five tested ratios. By incrementally moving the laser beam and detector across the flame, the study achieved fine mapping of temperature and OH concentration profiles, resolving OH peaks at sub-millimeter precision and capturing detailed flame chemistry structure.

CHAPTER 3: NUMERICAL METHOD

3.1. Principle of BAS

The theoretical basis of BAS starts with the Beer-Lambert law, which defines the theoretical absorbance, $\alpha(\nu)$, of light passing through a medium

$$\begin{aligned}\alpha(\nu) &= -\ln \left[\frac{I_t(\nu)}{I_0(\nu)} \right] = \sum_i [H_i \cdot \varphi_i(\nu)] \\ &= PXL \cdot \sum_i [S_{P,i}(T) \cdot \varphi_i(\nu)]\end{aligned}\quad (1)$$

where ν is frequency, $\varphi(\nu)$ is the area-normalized lineshape function, P is static pressure, X is mole fraction, L is optical path length, $S_P(T)$ is per-unit-pressure line-strength that is a function of temperature, $H = PXL \cdot S_P(T)$ is the integrated absorbance, and the subscript i signifies a specified transition.

However, the absorbance obtained in BAS differs from $\alpha(\nu)$ in Eq. (1). Both the incident intensity $I_0(\nu)$ and the transmitted intensity $I_t(\nu)$ are blurred by the spectrometer, as they represent convolutions of the true spectrum $I(\nu)$ with the instrument's response function $\Phi_i(\nu)$. This convolution causes the measured absorbance to appear broadened, a phenomenon known as instrument broadening. To distinguish it from α in Eq. (1), the BAS-measured absorbance is denoted as α_I and is expressed as

$$\begin{aligned}\alpha_I &= -\ln \left[\frac{I_t \otimes \Phi_I}{I_0 \otimes \Phi_I} \right]_\nu \\ &\approx -\ln \left\{ \frac{I_{f+l} - I_f}{I_l - I_b} \right\}\end{aligned}\quad (2)$$

where I_{f+l} , I_f , I_l , and I_b are four measured spectra. I_{f+l} is acquired with both flame and lamp on, while I_f is recorded with the flame on and the lamp off. Flame emission is subtracted

from I_{f+l} as $I_l \otimes \Phi_l = I_{f+l} - I_f$. I_l is obtained with only the lamp on, and I_b is the background subtracted from I_l as $I_0 \otimes \Phi_l = I_l - I_b$.

To determine T and X from $\alpha_l(\nu)$, the line strength $S_p(T)$ must be known. Parameters from the LIFBASE [3] and HITRAN [4] databases were employed to calculate $S_p(T)$ as

$$S_p(T) = \frac{Ac^2 g_n(2J_l+1)}{8\pi\nu_0^2 kTQ(T)} \exp\left(-\frac{E_u}{kT}\right) \left[1 - \exp\left(-\frac{h\nu_0}{kT}\right)\right] \quad (3)$$

where subscript u denotes the upper state, subscript l the lower state, h is Planck's constant, c the speed of light, k Boltzmann's constant, A the Einstein coefficient of radiation, g_n the nuclear statistical weight, J the rotational quantum number, $Q(T)$ the temperature-dependent partition function, ν_0 the transition's central frequency, and E as the corresponding state's energy. Since ν_0 in the UV range is very high ($>10^{15}$ Hz), the spontaneous emission term can be neglected with an error below 10^{-8} .

3.2 Partition function evaluation

For the OH radical at temperature T , the total partition function is the product of its components

$$Q_{tot}(T) = Q_{elec}(T) \cdot Q_{vib}(T) \cdot Q_{rot}(T) \quad (4)$$

where $Q_{tot}(T)$ represents the total partition function, $Q_{elec}(T)$ is the electronic partition function (accounts for different electronic states), $Q_{vib}(T)$ is the vibrational partition function (accounts for $v = 0, 1, 2, \dots$ states) and $Q_{rot}(T)$ is the rotational partition function (accounts for all J levels within each vibrational state).

The electronic partition function $Q_{elec}(T)$ is calculated by summing the degeneracy (g_e) of each electronic state, weighted by the Boltzmann factor related to that state's excitation energy (T_e)

$$Q_{elec}(T) = \sum_e \left[g_e \cdot \exp \left(-\frac{hcT_e}{k_B T} \right) \right] \quad (5)$$

The vibrational partition function $Q_{vib}(T)$ is calculated by summing the contributions of each vibrational level (v), weighted by the Boltzmann factor corresponding to the vibrational term ($E_{vib}(v)$)

$$Q_{vib}(T) = \sum_v \left[\exp \left(-\frac{hcE_{vib}(v)}{k_B T} \right) \right] \quad (6)$$

The term value $E_{vib}(v)$ is calculated using the harmonic and anharmonic constants provided in the diatomic constants table

$$E_{vib}(v) = \omega_e \left(v + \frac{1}{2} \right) - \omega_e x_e \left(v + \frac{1}{2} \right)^2 - E_{vib}(0) \quad (7)$$

where $E_{vib}(0)$ is the zero-point energy, which is subtracted so that the $v = 0$ level is the reference point ($E_{vib}(0) = 0$) for the summation.

The rotational partition function $Q_{rot}(T)$ is calculated by summing the degeneracy (g_J) of each rotational level, weighted by the Boltzmann factor corresponding to that level's rotational energy (E_{rot})

$$Q_{rot}(T) = \sum_{J''} \left[g_{J''} \cdot \exp \left(-\frac{hcE_{rot}(J'')}{k_B T} \right) \right] \quad (8)$$

The degeneracy g_J for an ideal rotational level J'' is typically $2J'' + 1$. The term value $E_{rot}(J'')$ is calculated using a second-order approximation for a diatomic molecule, including the correction for centrifugal distortion

$$E_{rot}(J'') = B_v J''(J'' + 1) - D_v (J''(J'' + 1))^2 \quad (9)$$

The rotational constant B_v decreases as the vibrational quantum number v increases due to centrifugal stretching of the bond

$$B_v = B_e - \alpha_e \left(v + \frac{1}{2} \right) \quad (10)$$

The centrifugal distortion constant D_v also changes slightly with v as

$$D_v = D_e + \beta_e \left(v + \frac{1}{2} \right) \quad (11)$$

3.3 Boltzmann line pair analysis for temperature measurement

Temperature can be determined from the relative population of rotational energy levels using Boltzmann analysis applied separately to absorption and emission spectra. After baseline correction and integration of individual OH spectral lines, each line's area can be related to the lower- or upper-state population through known spectroscopic parameters. For absorption, the integrated absorbance is proportional to the population of the lower state; for emission, the integrated line intensity is proportional to the population of the upper state. Plotting the logarithmic form of these relationships against the corresponding state energies yields a straight line whose slope is directly related to the flame temperature. This method allows temperature estimation without absolute radiometric calibration, relying instead on relative line strengths and known molecular parameters (Einstein coefficients, degeneracies, and energies).

3.3.1 Absorption Boltzmann Relation

Using the line area W and wavenumber $\bar{\nu}$

$$\ln \left(\frac{W \bar{\nu}^2}{g'' A_{ul}} \right) = - \frac{E''}{k_B T} + C \quad (12)$$

3.3.2 Emission Boltzmann Relation

Using integrated emission intensity I

$$\ln \left(\frac{I}{g' A_{ul} \bar{\nu}} \right) = - \frac{E'}{k_B T} + C \quad (13)$$

3.3.3 Temperature Extraction

For both absorption and emission, a Boltzmann plot constructed between the state energy E'' (absorption) or E' (emission) vs the logarithmic expression above yields a straight line, with the expression for calculating the slope as

$$\text{slope} = -\frac{1}{k_B T} \Rightarrow T = -\frac{1}{k_B \cdot \text{slope}} \quad (14)$$

3.4 Line-of-sight absorption

The radial distributions of temperature $T(r)$ and mole fraction $X(r)$ can be converted into single representative values by simulating line-of-sight absorption. The line-of-sight absorption calculation follows the well-known Abel transform. When a light beam traverses an axisymmetric medium, the average absorbance can be expressed as

$$\alpha = 2 \int_d^R \frac{k(r).r}{\sqrt{r^2-d^2}} dr \quad (12)$$

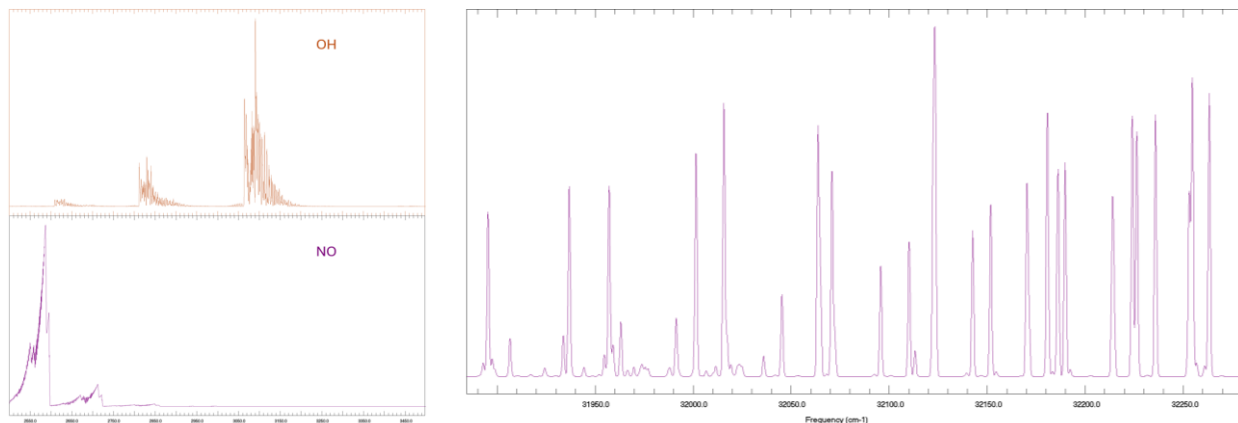
where d is the distance from the beam to the axis, and $k(r)$ is the absorption coefficient, which depends on $T(r)$ and $X(r)$ given by

$$k(r) = P.X(r).\sum_i [S_{P,i}\{T(r)\}.\varphi_i] \quad (13)$$

where subscript i denotes individual transitions.

CHAPTER 4: EXPERIMENTAL SETUP

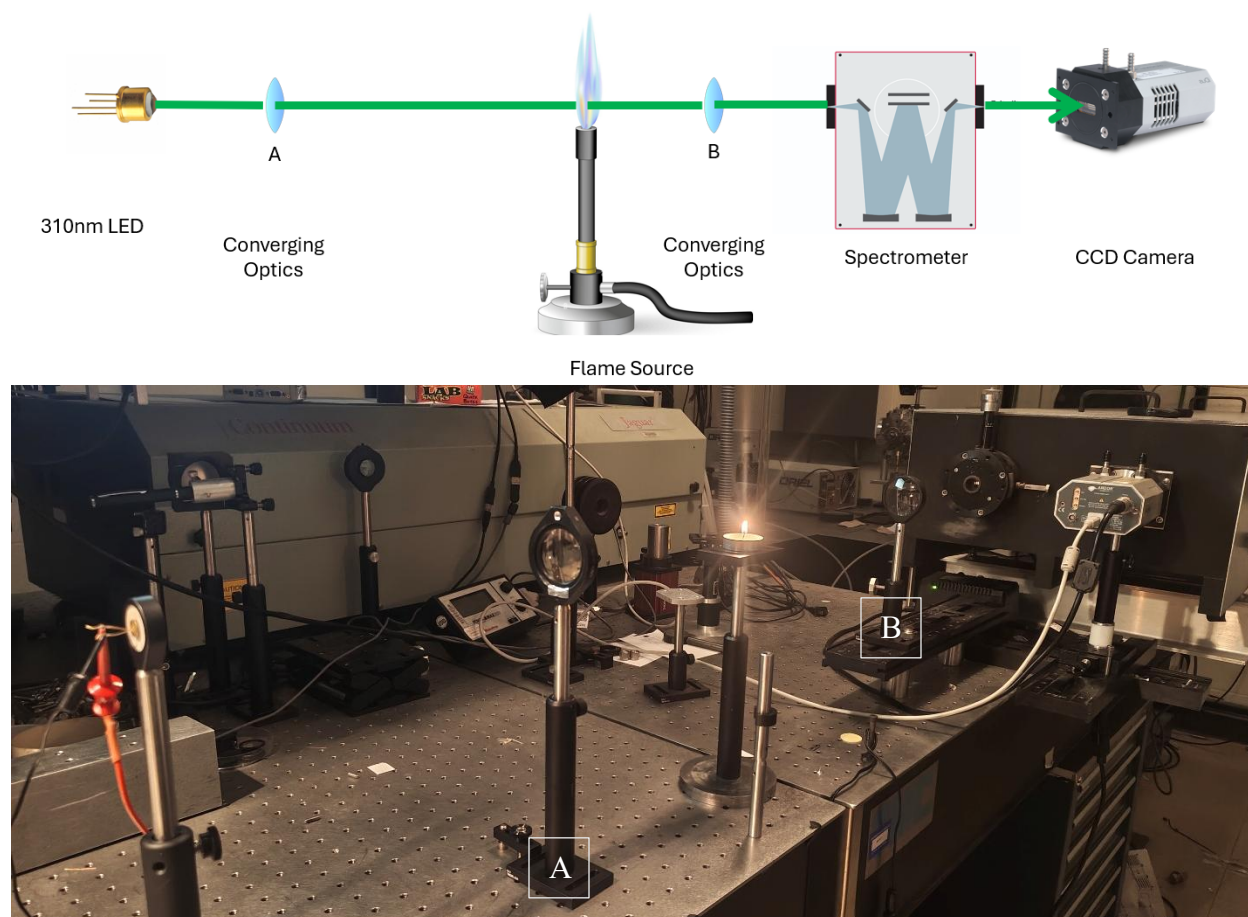
The 0-0 band of OH between 311 and 313.5 nm was chosen due to it having the strongest absorption among the OH transitions and minimal interference from other combustion radicals, as shown in Figs. 1 & 2.



Figs. 1 & 2: OH and NO line-strengths at 1800 K. Data from LIFBASE 2.1.3 (left); OH line-strengths between 311 and 313.5 nm (right)

The schematic and the actual setup for the measurements in Figs. 3 & 4 show the flame and the measurement system. The BAS measurement system consisted of a light source, a spectrometer, and an optical path arrangement. Broadband radiation spanning 311-313.5 nm was produced by a 310nm LED as the laser source, which gives a good signal and excellent spatial resolution (~ 1 mm). This light was then collimated into a narrow beam using the optical arrangement. A and B were quartz-glass convex lenses with focal lengths of 29.21 cm and 24.13 cm, respectively. A wavelength calibration was performed using an Hg lamp to ensure accurate spectral alignment. The beam was first positioned to traverse through the flame at a height of 0.25" above the source. After traversing the flame, the beam was collected and directed into a grating spectrometer (1.54m Czerny-Turner spectrometer with 9-degree deviation angle) equipped with a 2400 grooves/mm grating and a CCD camera (Andor iDus DU420 CCD detector featuring a 256×1024 -pixel array).

with 25-micron square pixels, ensuring high-resolution (0.012 nm range) spectral acquisition). The spectrometer dispersed the broadband light into a spectrum that was recorded by the CCD. Each spectral frame was accumulated over 100 ms, and 20 frames were averaged to reduce noise, requiring approximately 2 s per spectrum. Spectra were collected at 2 heights above the source (0.25" and 0.5"), to map the OH concentration and temperature profiles.



Figs. 3 & 4: Outline of the experimental setup, highlighting various components and the path traced by the light from the source to the detector (top); snapshot of the actual setup with the candle flame (bottom)

To isolate the molecular absorption (or emission) features from the slowly varying broadband background, a rolling-percentile masking method was used. First, a high percentile (typically 90-98%) of the signal was computed within a sliding window across the spectrum, producing an adaptive upper envelope that follows only the continuum and not the narrow spectral lines. Points lying significantly below this envelope (for absorption) or above it (for emission), based on a noise-

derived threshold, were automatically identified as spectral-line regions and excluded from the baseline fit. A low-order polynomial (2nd order) was then fitted only to the unmasked continuum points. Finally, the baseline was removed from the raw spectrum by division for absorption (to obtain I/I_0 with a flat baseline) or subtraction for emission to yield a clean, continuum-corrected spectrum.

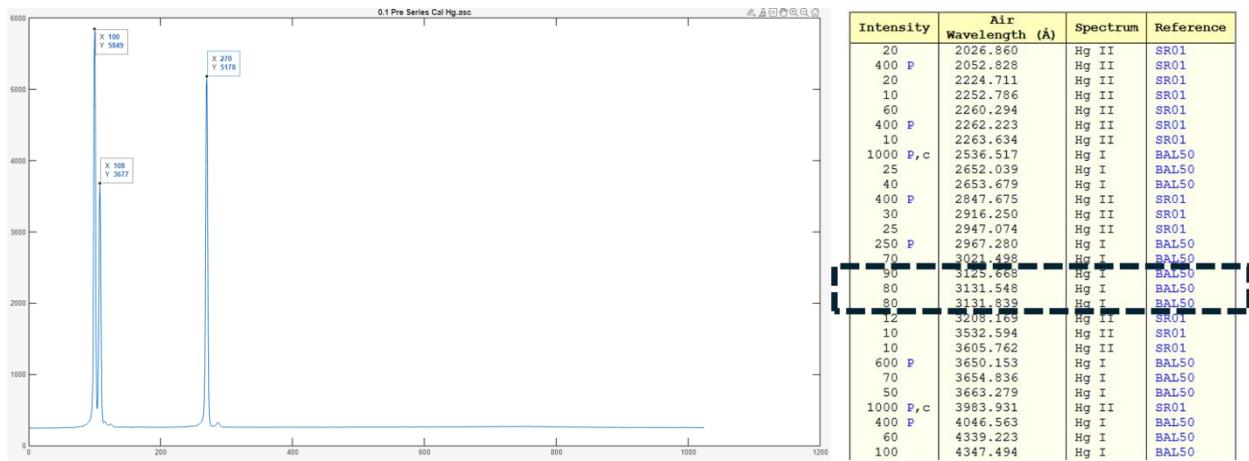
CHAPTER 5: RESULTS & DISCUSSION

5.1. Wavelength calibration using a mercury (Hg) lamp

Wavelength calibration of the spectrometer was carried out using a low-pressure mercury (Hg) discharge lamp, which provides several narrow, well-characterized emission lines suitable for establishing a pixel–wavelength relationship. In the spectral region relevant to the OH A-X band, three distinct Hg emission peaks were clearly observed, as shown in Fig. 5, and their corresponding pixel positions were extracted from the recorded spectrum. These peaks were then matched to their known wavelengths from NIST reference tables, as shown in Fig. 6. Using the three pixel–wavelength pairs, a quadratic calibration curve of the form

$$\lambda = a_0 + a_1p + a_2p^2 \quad (14)$$

where p is the pixel number, which was fitted to account for the small non-linearity inherent in the Czerny–Turner optical dispersion. The resulting polynomial provided a smooth and accurate mapping from pixel space to physical wavelength, and this calibration was subsequently applied to all absorption and emission measurements.

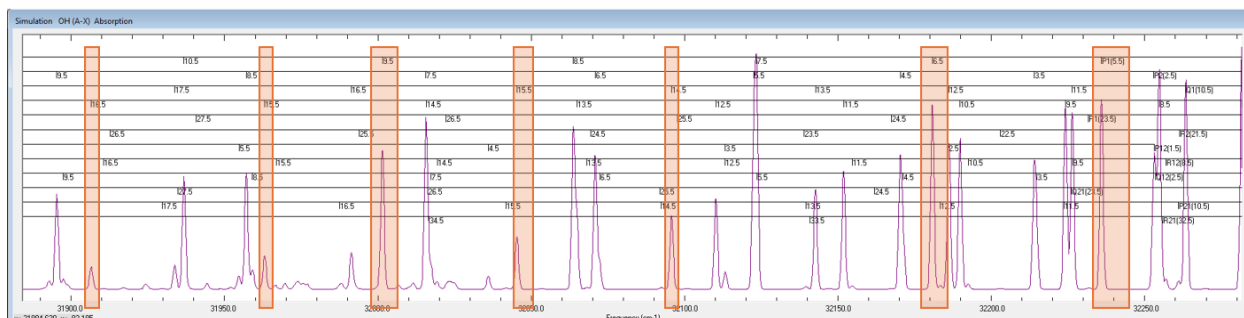


Figs. 5 & 6: Hg emission spectrum (left); strong lines of Hg I from NIST (right)

5.2. Theoretical OH absorption spectrum and extraction of line parameters

The theoretical OH A-X (0-0) absorption spectrum was generated using LIFBASE, as shown in Fig. 7, which provides spectroscopic constants and line-by-line transition data for the OH radical. The software was configured for the desired rotational band system, and individual transitions corresponding to the experimentally observed P1, Q1, and Q2 branches were identified. For each selected line, LIFBASE outputs the line position (wavelength in air and wavenumber), Einstein A_{ul} coefficient, and the corresponding rotational quantum numbers. From these, the lower-state energy E'' and the degeneracy term $g'' = 2J'' + 1$ were extracted using the HITRAN database, as shown in Table 1, for use in the Boltzmann analysis.

Only well-resolved, isolated OH transitions were chosen to avoid peak overlap and baseline interference. Lines were selected to span a sufficiently wide range of lower-state energies E'' , ensuring good sensitivity to temperature in the Boltzmann plot. Additionally, transitions with reliable LIFBASE line strengths and minimal spectral interference from broadband flame emission or nearby interfering species were prioritized. The selected lines are highlighted in Fig. 7. The theoretical line strengths provided by LIFBASE served as the temperature-independent absorption coefficients needed for relating measured equivalent widths to the population of the lower rotational levels. These collected spectroscopic parameters formed the basis for constructing Boltzmann plots and determining the flame temperature from measured OH absorption features.



Line Number (J'')	Lambda (air) (Å)	Wavenumber (cm ⁻¹)	E'' (cm ⁻¹)	Aul (s ⁻¹)	g'
P1 (5.5)	3101.23	32235.91	543.5955	4.324e5	4.5
P1 (6.5)	3106.55	32180.76	767.4816	4.151e5	5.5
Q1 (14.5)	3114.78	32095.73	3819.1203	5.837e5	14.5
Q1 (15.5)	3119.68	32045.32	4351.2582	5.683e5	15.5
P1 (9.5)	3123.95	32001.5	1650.8104	3.767e5	8.5
Q2 (15.5)	3127.70	31963.09	4939.7714	5.491e5	15.5
Q2 (16.5)	3133.25	31906.54	5533.4736	5.317e5	16.5

Table 1: Selected lines and parameters extracted using LIFBASE and HITRAN

5.3. Absorption spectrum analysis of butane flame

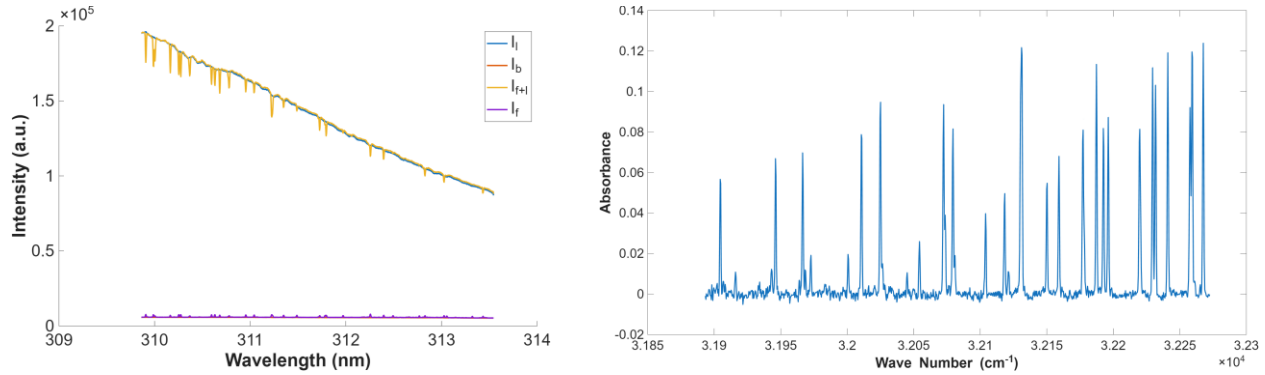


Fig. 8: Butane flame torch and the two laser paths

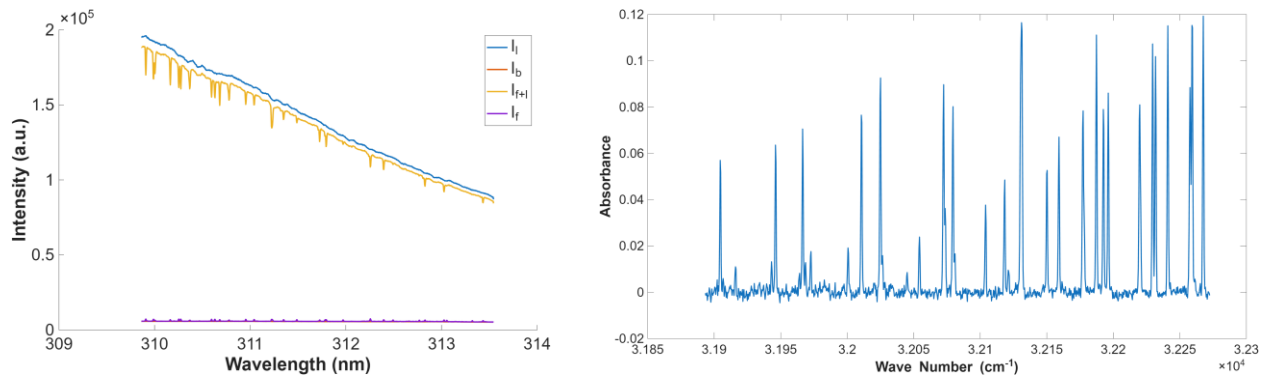
For the butane flame, absorption measurements were performed at two vertical locations above the burner. The four required spectra I_{f+l} , I_f , I_l , and I_b were acquired as shown in Figs. 9 and 11. Following zero-baseline correction and wavelength calibration, the absorbance spectra were computed (Figs. 10 and 12), and multiple isolated OH A-X (0-0) rotational lines were integrated to obtain their equivalent widths. These values were then used to construct the Boltzmann plots for each measurement height (Figs. 13 and 14). From the slope of each plot, the

rotational temperature was extracted and subsequently used to determine the OH column density via the line-strength formulation. With the known optical path length and the ideal-gas relation, the column density was converted into a local OH mole fraction.

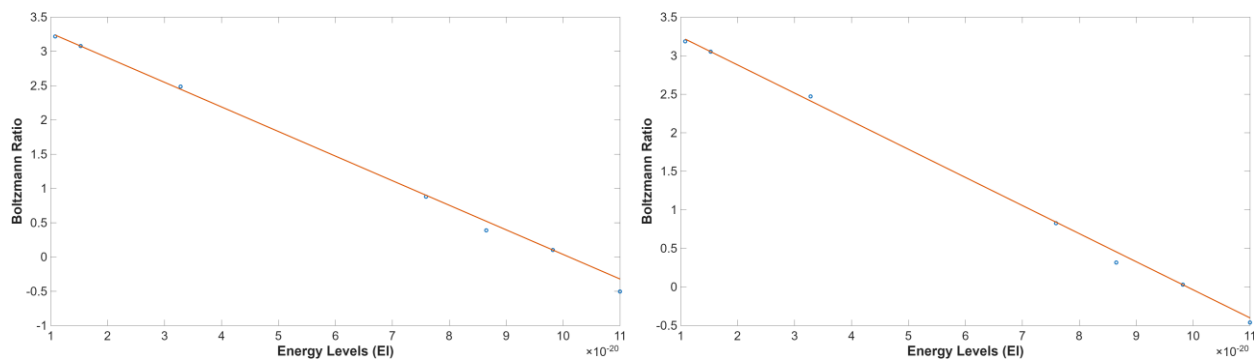
At the lower sampling height of 0.25", the fitted temperature was 2020 K, reflecting the hotter region near the primary reaction zone where OH production is most intense. The corresponding OH mole fraction, 0.0167, was also highest at this location. At the upper position (0.5"), both the temperature (1988 K) and mole fraction (0.0162) decreased, consistent with cooling and dilution as combustion products mix with the surrounding air. The linearity of the Boltzmann plots at both locations confirms that OH rotational populations remain near equilibrium, supporting the validity of the absorption-based method.



Figs. 9 & 10: Spectrum intensities measured (left) and absorbance data (right) at the flame height of 0.25"



Figs. 11 & 12: Spectrum intensities measured (left) and absorbance data (right) at the flame height of 0.5"



Figs. 13 & 14: Boltzmann plots at flame height of 0.25" (left) and 0.5" (right)

5.4. Absorption spectrum analysis of candle flame

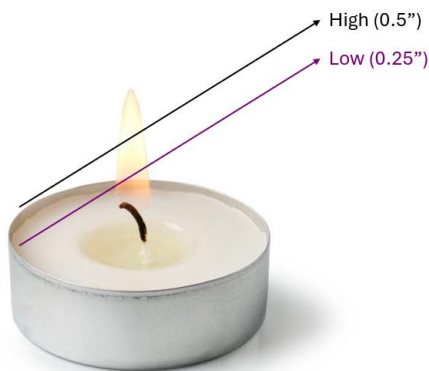
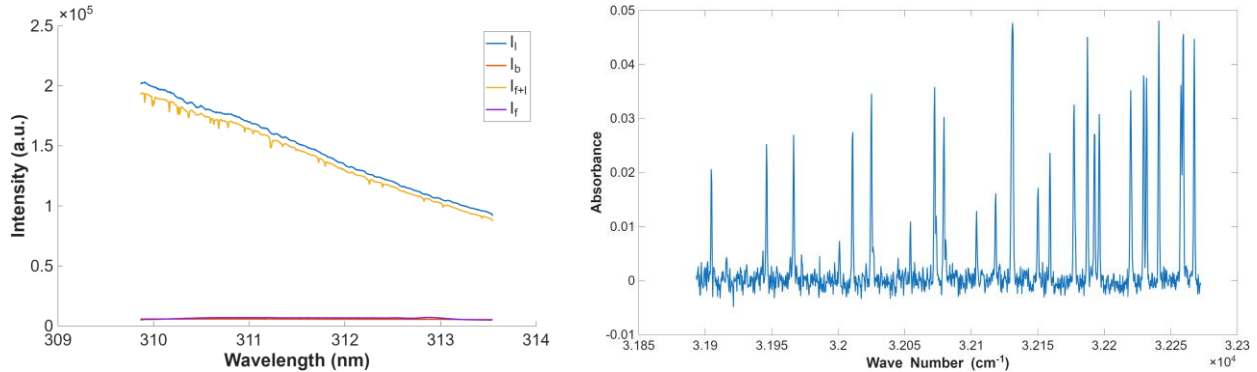


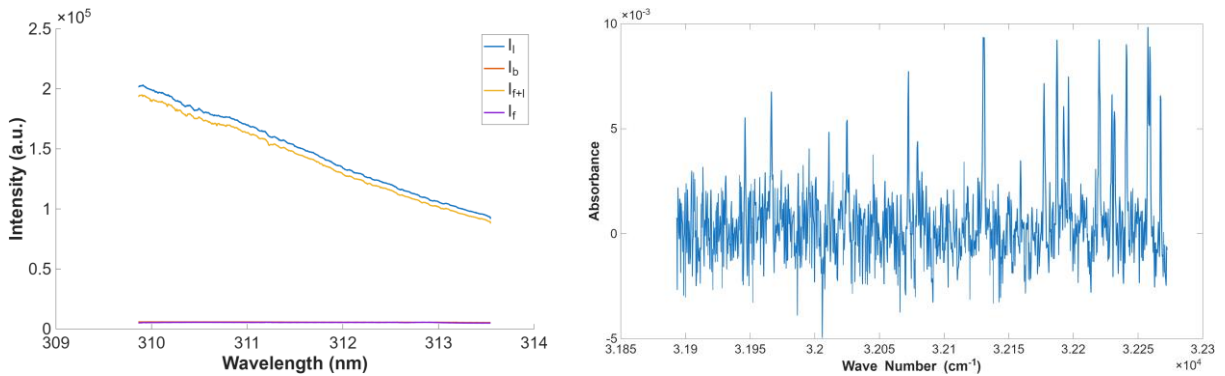
Fig. 15: Candle flame and the two laser paths

For the candle flame, absorption measurements were performed at two vertical locations above the burner. The four required spectra I_{f+l} , I_f , I_l , and I_b were acquired as shown in Figs. 16 and 18. Following zero-baseline correction and wavelength calibration, the absorbance spectra were computed (Figs. 17 and 19), and multiple isolated OH A-X (0-0) rotational lines were integrated to obtain their equivalent widths. These values were then used to construct the Boltzmann plots for each measurement height (Figs. 20 and 21). From the slope of each plot, the rotational temperature was extracted and subsequently used to determine the OH column density via the line-strength formulation. With the known optical path length and the ideal-gas relation, the column density was converted into a local OH mole fraction.

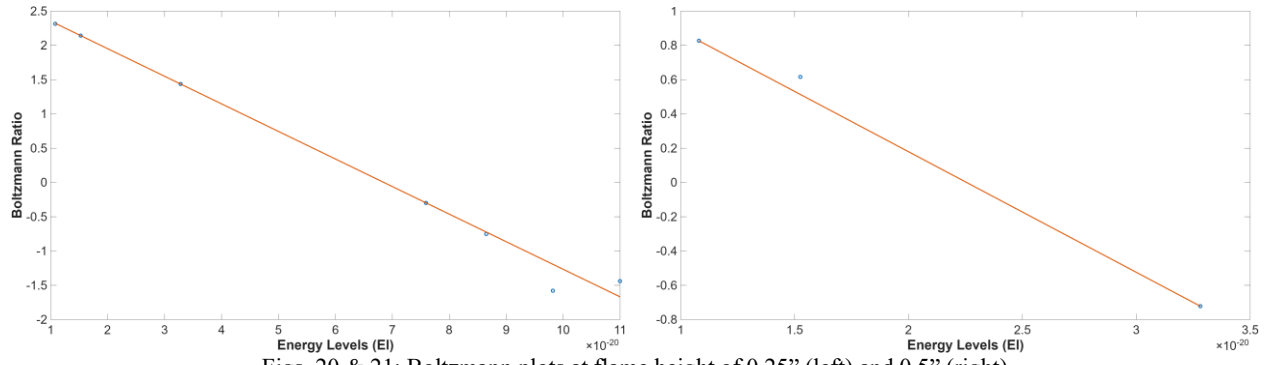
At the lower sampling height of 0.25'' in the candle flame, the Boltzmann fit yielded a temperature of approximately 1810 K, which is consistent with the cooler, diffusion-controlled nature of wax combustion compared to gaseous fuels. The corresponding OH mole fraction was significantly lower (0.0056), reflecting the weaker production of OH radicals in hydrocarbon-rich, sooting flames. At the higher measurement position (0.5''), both the fitted temperature (1030 K) and mole fraction (0.00056) decreased further, indicating dilution by entrained air and partial quenching of radicals as the plume cools. Unlike gaseous fuels, the candle flame exhibited a higher continuum background due to soot luminosity, but the linearity of the Boltzmann plot remained adequate after baseline correction. This confirms that, despite weaker OH features, the rotational population is sufficiently close to equilibrium for reliable temperature and concentration extraction.



Figs. 16 & 17: Spectrum intensities measured (left) and absorbance data (right) at the flame height of 0.25''



Figs. 18 & 19: Spectrum intensities measured (left) and absorbance data (right) at the flame height of 0.5''



Figs. 20 & 21: Boltzmann plots at flame height of 0.25'' (left) and 0.5'' (right)

5.5. Absorption spectrum analysis of sterno flame

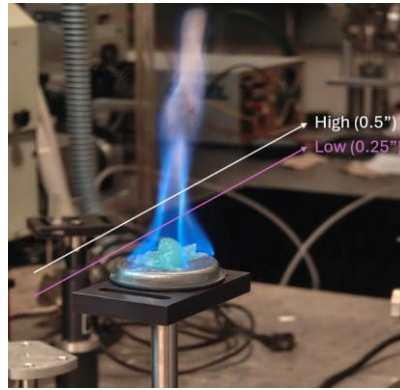
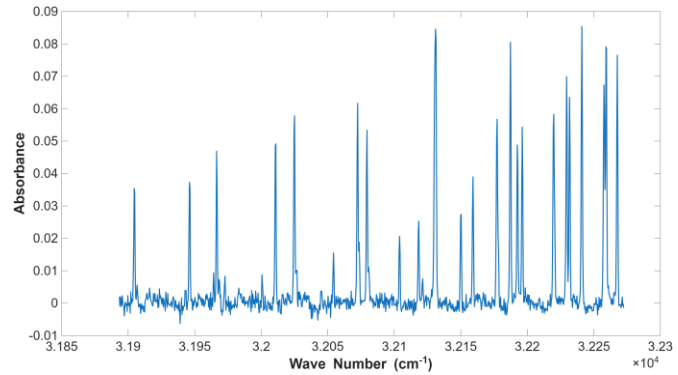
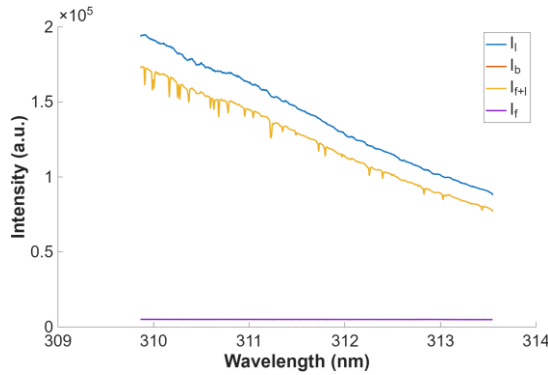


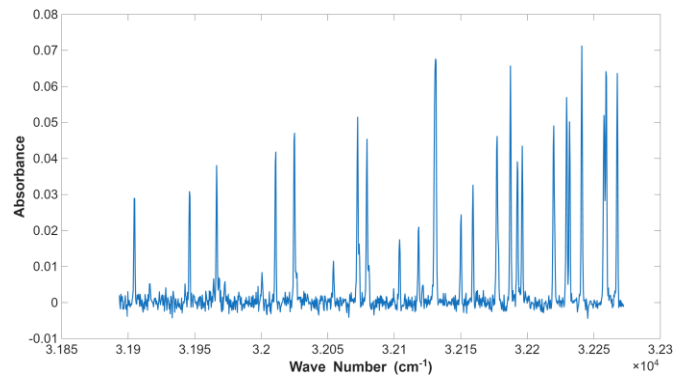
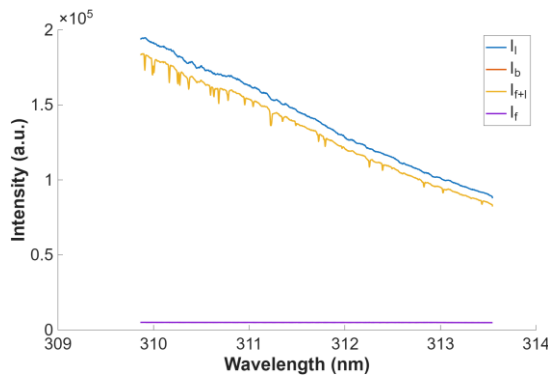
Fig. 22: Sterno flame and the two laser paths

For the sterno flame, absorption measurements were performed at two vertical locations above the burner. The four required spectra I_{f+l} , I_f , I_l , and I_b were acquired as shown in Figs. 23 and 25. Following zero-baseline correction and wavelength calibration, the absorbance spectra were computed (Figs. 24 and 26), and multiple isolated OH A-X (0-0) rotational lines were integrated to obtain their equivalent widths. These values were then used to construct the Boltzmann plots for each measurement height (Figs. 27 and 28). From the slope of each plot, the rotational temperature was extracted and subsequently used to determine the OH column density via the line-strength formulation. With the known optical path length and the ideal-gas relation, the column density was converted into a local OH mole fraction.

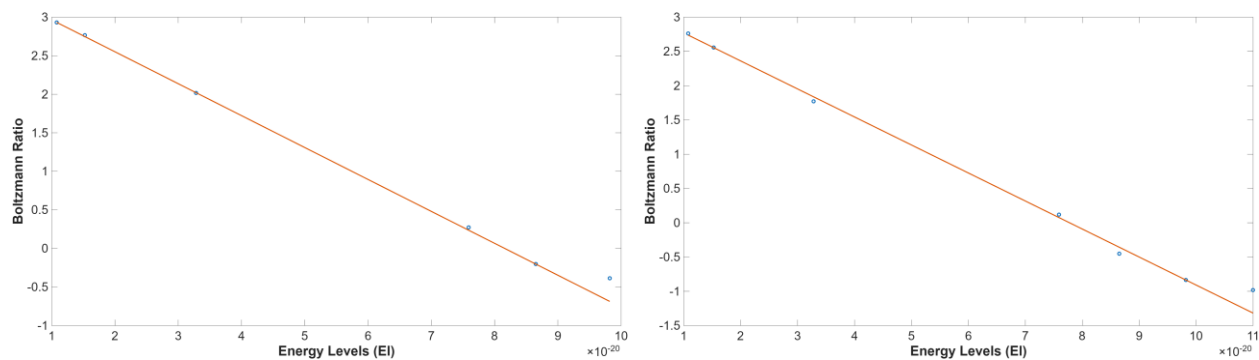
At the lower sampling height of 0.25'' in the Sterno flame, the Boltzmann fit yielded a temperature of approximately 1773 K, consistent with the cooler, alcohol-based combustion chemistry of gelled ethanol/methanol fuels. The corresponding OH mole fraction was 0.0088, reflecting moderate OH production typical of clean, low-soot flames with relatively low heat release. At the higher measurement position (0.5''), both the fitted temperature (1746 K) and OH mole fraction (0.0083) decreased, driven by enhanced entrainment of ambient air and progressive radical recombination as the plume cools. Unlike hydrocarbon flames, the Sterno flame exhibited minimal continuum background because of its essentially soot-free combustion, improving the signal-to-noise ratio of the OH absorption features. After baseline correction, the Boltzmann plot retained strong linearity, confirming that the rotational population remains sufficiently equilibrated for reliable extraction of temperature and OH concentration across the sampled heights.



Figs. 23 & 24: Spectrum intensities measured (left) and absorbance data (right) at the flame height of 0.25''



Figs. 25 & 26: Spectrum intensities measured (left) and absorbance data (right) at the flame height of 0.5''



Figs. 27 & 28: Boltzmann plots at flame height of 0.25" (left) and 0.5" (right)

5.6. Absorption spectrum analysis of hexamine flame

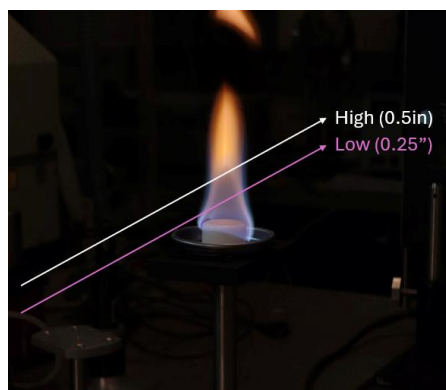
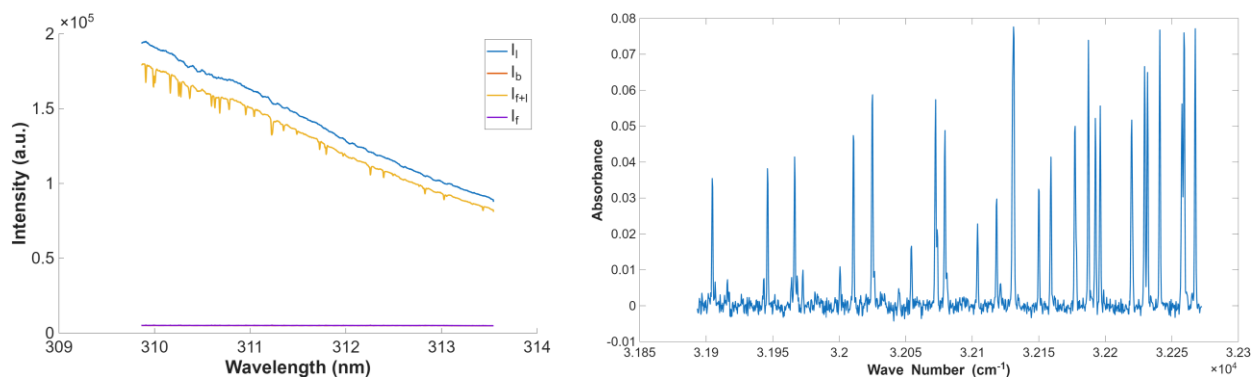


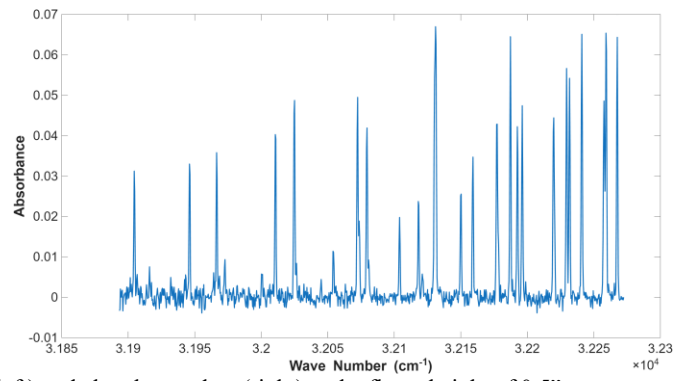
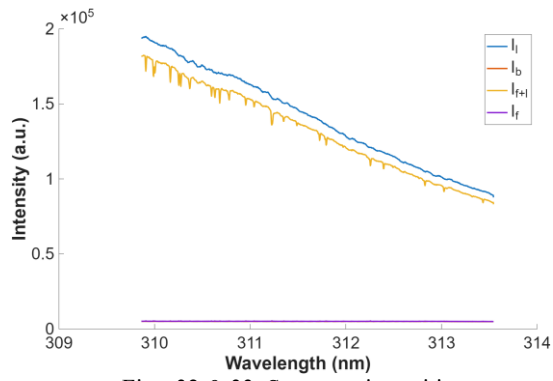
Fig. 29: Hexamine flame and the two laser paths

For the hexamine flame, absorption measurements were performed at two vertical locations above the burner. The four required spectra I_{f+l} , I_f , I_l , and I_b were acquired as shown in Figs. 30 and 32. Following zero-baseline correction and wavelength calibration, the absorbance spectra were computed (Figs. 31 and 33), and multiple isolated OH A-X (0-0) rotational lines were integrated to obtain their equivalent widths. These values were then used to construct the Boltzmann plots for each measurement height (Figs. 34 and 35). From the slope of each plot, the rotational temperature was extracted and subsequently used to determine the OH column density via the line-strength formulation. With the known optical path length and the ideal-gas relation, the column density was converted into a local OH mole fraction.

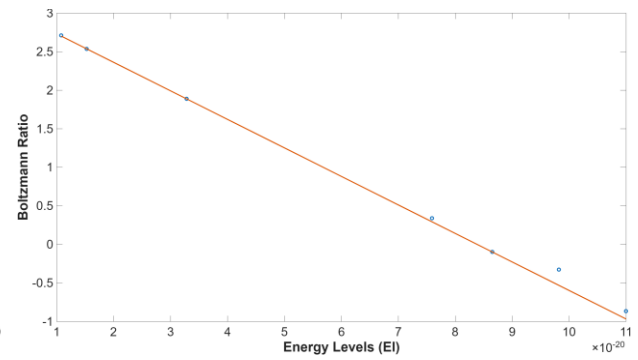
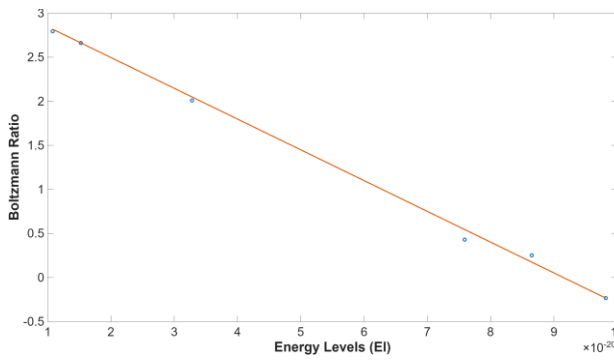
At the lower sampling height of 0.25'' in the hexamine flame, the Boltzmann fit yielded a temperature of approximately 2075 K, consistent with the relatively hot but strongly fuel-rich, nitrogen-containing combustion characteristic of solid hexamine tablets. The corresponding OH mole fraction was 0.0119, reflecting moderate OH formation in the luminous, partially sooting reaction zone where incomplete oxidation limits radical production. At the higher measurement location (0.5''), both the fitted temperature (1959 K) and OH mole fraction (0.0102) decreased, driven by entrainment-induced cooling and the recombination or conversion of OH as the plume transitions toward a diffusion-dominated post-flame region. The hexamine flame exhibited a noticeable broadband background due to soot and thermally excited CN/HCN chemiluminescence, requiring careful baseline correction before line selection and integration. After this correction, the Boltzmann plot maintained adequate linearity, confirming that the OH rotational population remained near equilibrium and permitting reliable extraction of temperature and local radical concentration.



Figs. 30 & 31: Spectrum intensities measured (left) and absorbance data (right) at the flame height of 0.25''



Figs. 32 & 33: Spectrum intensities measured (left) and absorbance data (right) at the flame height of 0.5"



Figs. 34 & 35: Boltzmann plots at flame height of 0.25" (left) and 0.5" (right)

5.7. Absorption spectrum analysis of propane flame

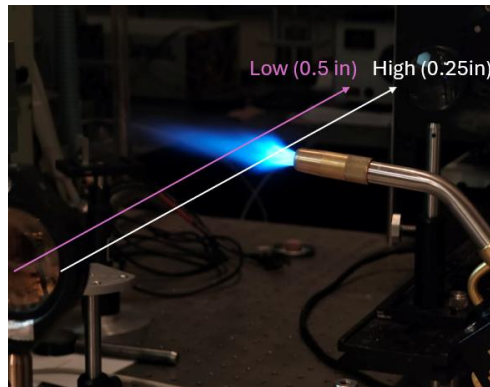
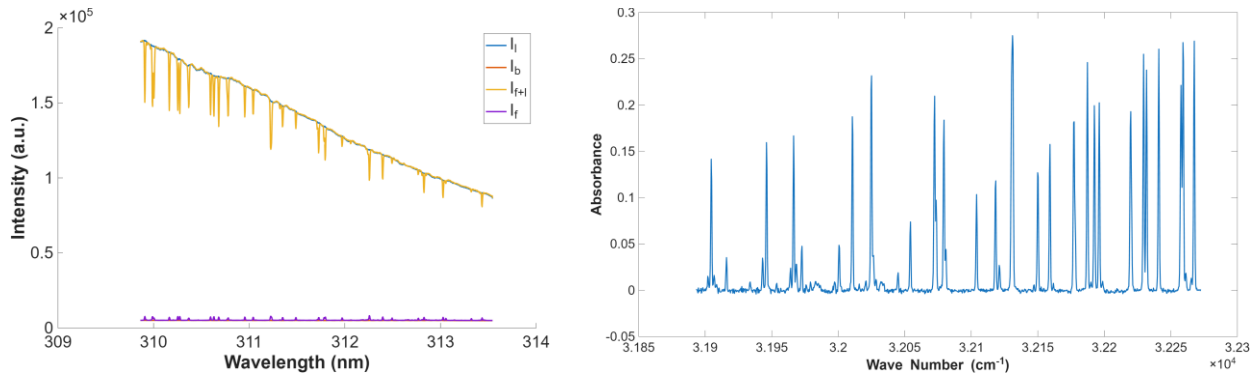


Fig. 36: Propane flame and the two laser paths

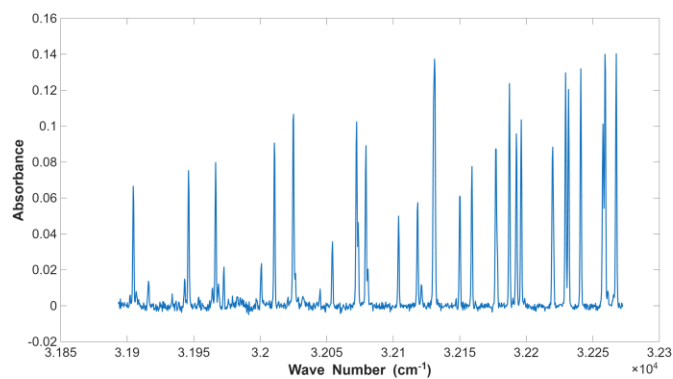
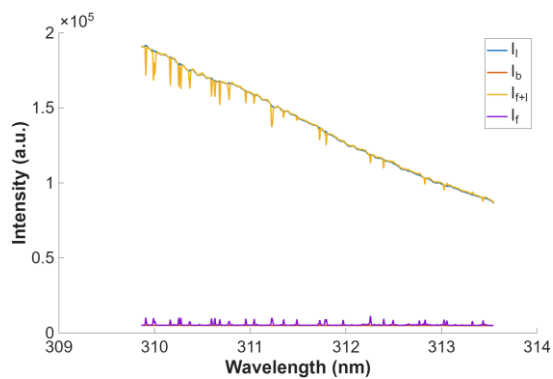
For the propane flame, absorption measurements were performed at two vertical locations above the burner. The four required spectra I_{f+l} , I_f , I_l , and I_b were acquired as shown in Figs. 37 and 39. Following zero-baseline correction and wavelength calibration, the absorbance spectra

were computed (Figs. 38 and 40), and multiple isolated OH A-X (0-0) rotational lines were integrated to obtain their equivalent widths. These values were then used to construct the Boltzmann plots for each measurement height (Figs. 41 and 42). From the slope of each plot, the rotational temperature was extracted and subsequently used to determine the OH column density via the line-strength formulation. With the known optical path length and the ideal-gas relation, the column density was converted into a local OH mole fraction.

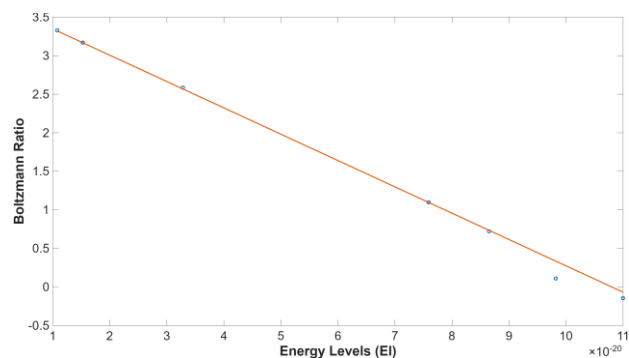
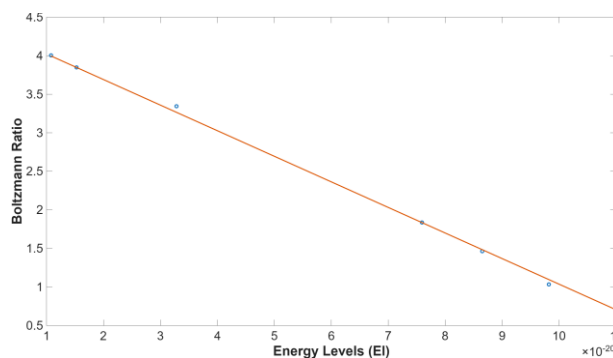
At the lower sampling height of 0.25” in the propane flame, the Boltzmann fit yielded a temperature of approximately 2184 K, consistent with propane’s clean, gaseous, premixed-like combustion behavior that generates strong OH formation near the primary reaction zone. The corresponding OH mole fraction was 0.043, reflecting the high radical production typical of blue, low-soot hydrocarbon flames. At the higher measurement position (0.5”), both the fitted temperature (2122 K) and OH mole fraction (0.02) decreased, as expected from post-flame cooling and dilution with entrained ambient air. Propane exhibited minimal broadband luminosity compared to solid-fuel flames, resulting in a clean baseline with well-isolated OH A-X (0-0) lines and highly linear Boltzmann plots. This linearity confirms that OH rotational populations remained close to equilibrium across both heights, enabling reliable extraction of local temperature and radical concentration.



Figs. 37 & 38: Spectrum intensities measured (left) and absorbance data (right) at the flame height of 0.25”



Figs. 39 & 40: Spectrum intensities measured (left) and absorbance data (right) at the flame height of 0.5''



Figs. 41 & 42: Boltzmann plots at flame height of 0.25'' (left) and 0.5'' (right)

5.8. Theoretical OH emission spectrum and extraction of line parameters

The theoretical OH A-X (0-0) emission spectrum was generated using LIFBASE [3] as shown in Fig. 43, which provides spectroscopic constants and line-by-line transition data for the OH radical. The software was configured for the desired rotational band system, and individual transitions corresponding to the experimentally observed P1, Q1, and Q2 branches were identified. For each selected line, LIFBASE outputs the line position (wavelength in air and wavenumber), Einstein A_{ul} coefficient, and the corresponding rotational quantum numbers. From these, the upper-state energy E' and the degeneracy term $g' = 2J' + 1$ were extracted by adding the photon energy to the lower state energy and by using the HITRAN database, resp., as shown in Table 1, for use in the Boltzmann analysis.

Only well-resolved, isolated OH transitions were chosen to avoid peak overlap and baseline interference. Lines were selected to span a sufficiently wide range of upper-state energies E' , ensuring good sensitivity to temperature in the Boltzmann plot. Additionally, transitions with reliable LIFBASE line strengths and minimal spectral interference from broadband flame emission or nearby interfering species were prioritized. The selected lines are highlighted in Fig. 43.

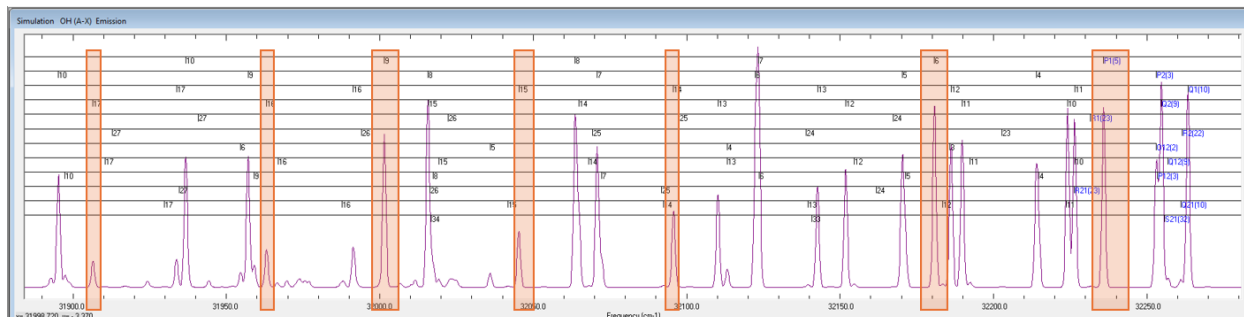
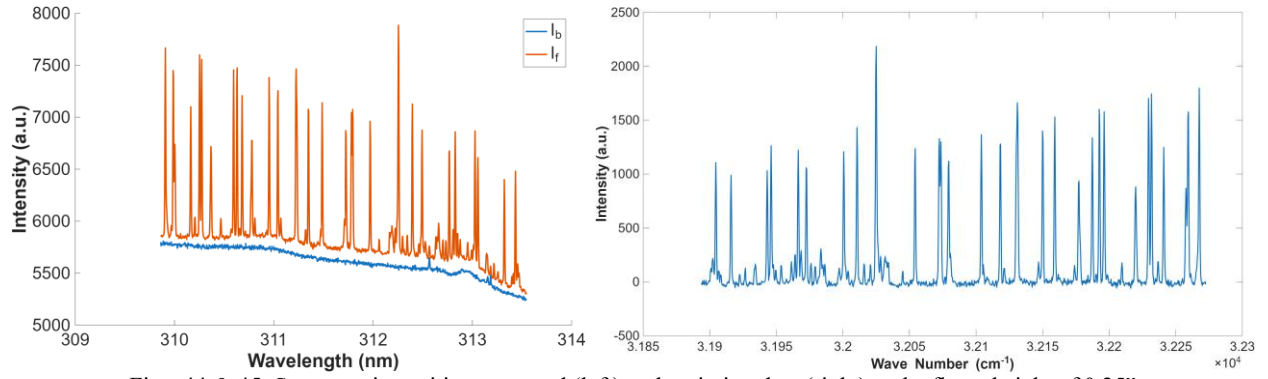


Fig. 43: Theoretical OH A-X 0-0 emission spectrum generated using LIFBASE

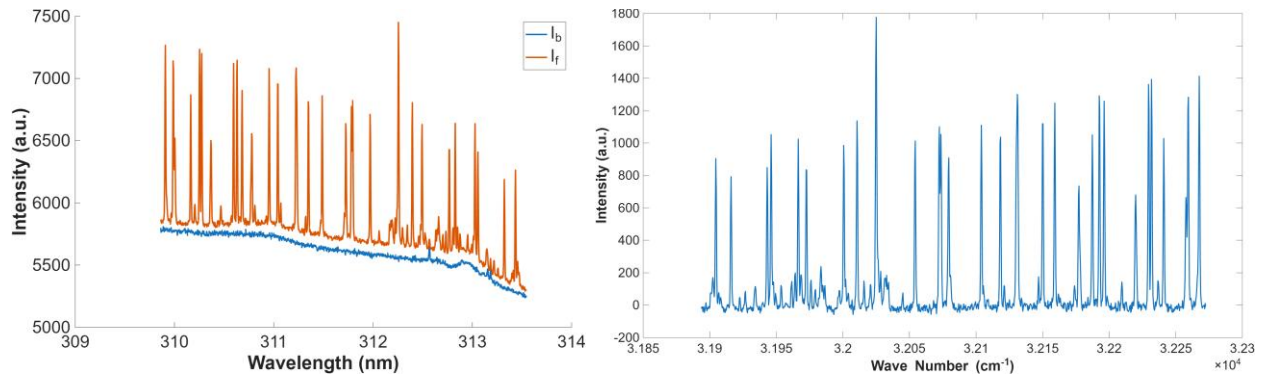
5.9. Emission spectrum analysis of butane flame

For the butane flame, emission measurements were performed at two vertical locations above the burner. The two required spectra I_f and I_b were acquired as shown in Figs. 44 and 46. Following zero-baseline correction and wavelength calibration, the emission spectra were computed (Figs. 45 and 47), and multiple isolated OH A-X (0-0) rotational lines were integrated to obtain their equivalent widths. These values were then used to construct the Boltzmann plots for each measurement height (Figs. 48 and 49). From the slope of each plot, the rotational temperature was extracted.

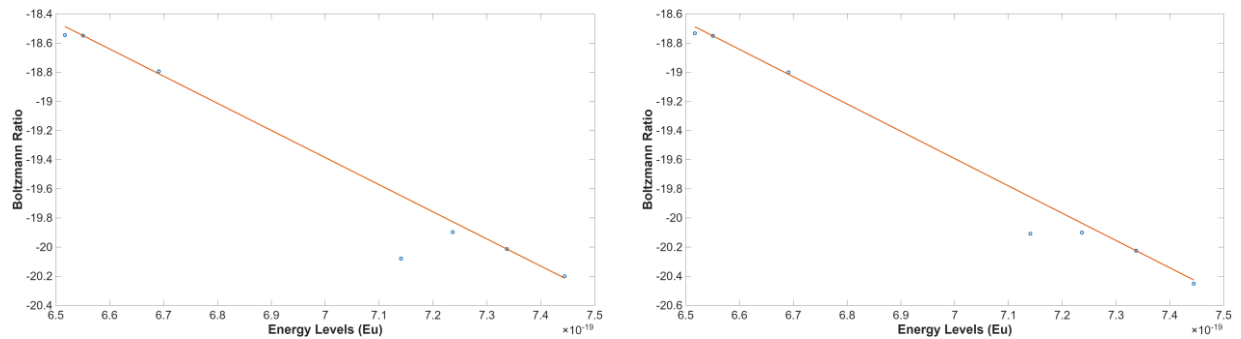
At the lower sampling height of 0.25" in the butane flame, the Boltzmann fit yielded a temperature of approximately 3902 K, while at the higher measurement position (0.5"), the fitted temperature (3865 K) decreased.



Figs. 44 & 45: Spectrum intensities measured (left) and emission data (right) at the flame height of 0.25''



Figs. 46 & 47: Spectrum intensities measured (left) and emission data (right) at the flame height of 0.5''

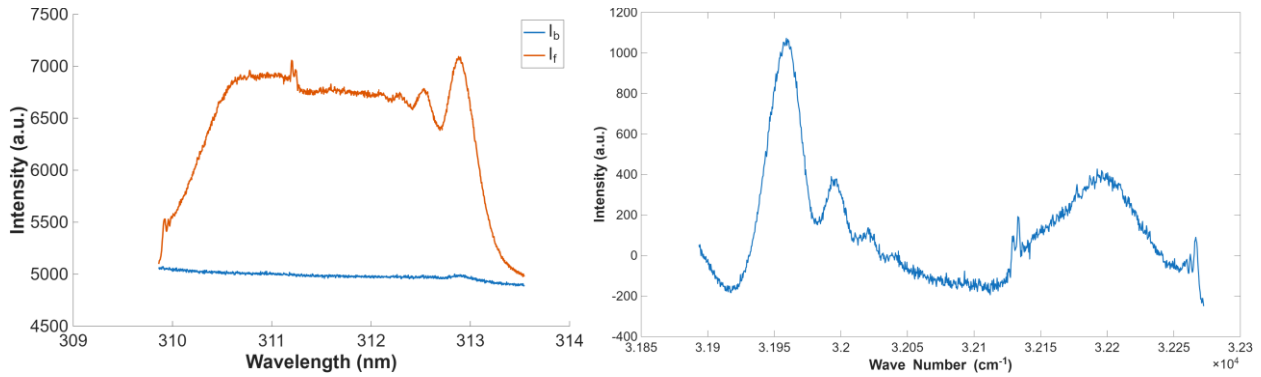


Figs. 48 & 49: Boltzmann plots at flame height of 0.25'' (left) and 0.5'' (right)

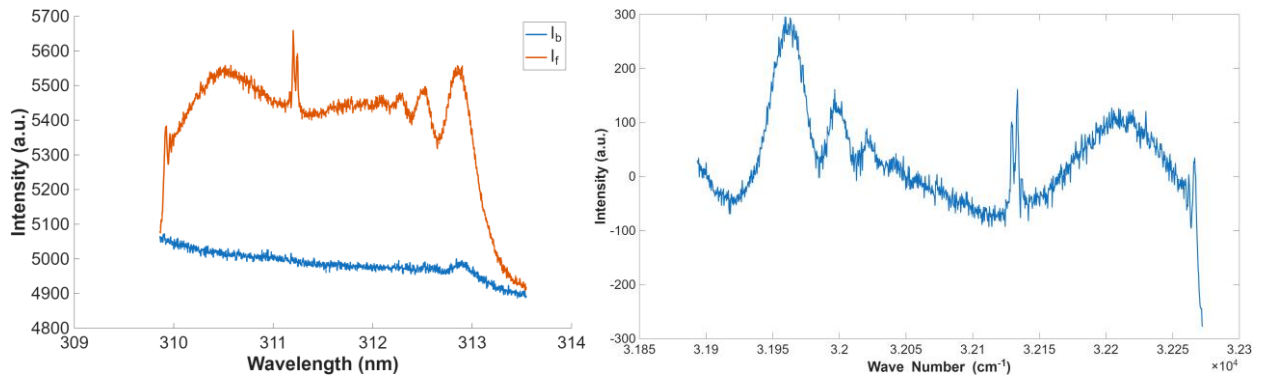
5.10. Emission spectrum analysis of candle flame

For the candle flame, emission measurements were performed at two vertical locations above the burner. The two required spectra I_f and I_b were acquired as shown in Figs. 50 and 52. Following zero-baseline correction and wavelength calibration, the emission spectra were computed (Figs. 51 and 53).

The recorded spectrum shows indistinct rotational line features, which could be attributed to the low intensity of the OH signal and the strong continuum background from the soot. Candle flames are highly luminous diffusion flames characterized by high soot formation in the yellow/orange zone. This soot radiates as a near-perfect blackbody emitter due to its high temperature. The blackbody radiation creates a very strong, broad, continuous background signal that underlies the specific OH emission peaks. The observed spectrum shows the intensity starting high (around 300-400 arbitrary units) even between the peaks, indicating a significant continuum signal. This high background drastically reduces the signal-to-noise ratio (SNR) and the contrast between the discrete OH rotational peaks and the baseline noise, making the individual OH lines look blurry or washed out.



Figs. 50 & 51: Spectrum intensities measured (left) and emission data (right) at the flame height of 0.25''

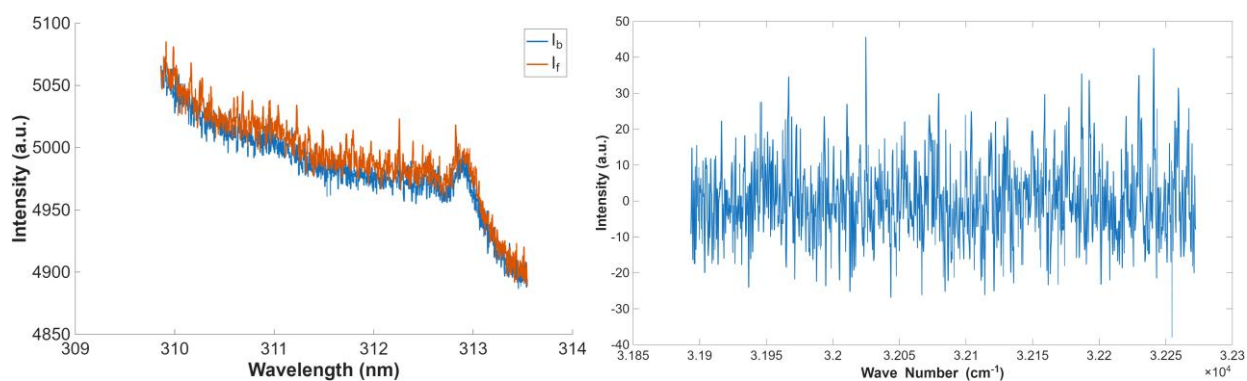


Figs. 52 & 53: Spectrum intensities measured (left) and emission data (right) at the flame height of 0.5''

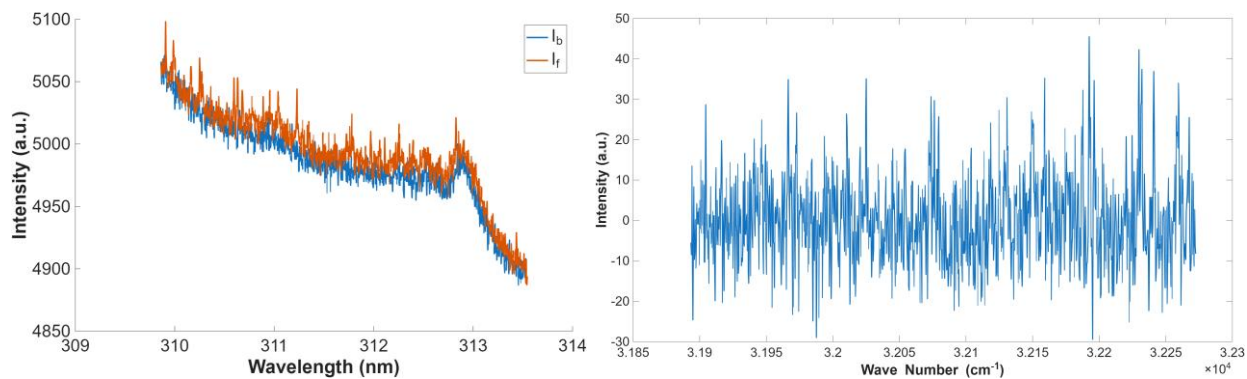
5.11. Emission spectrum analysis of sterno flame

For the sterno flame, emission measurements were performed at two vertical locations above the burner. The two required spectra I_f and I_b were acquired as shown in Figs. 54 and 56. Following zero-baseline correction and wavelength calibration, the emission spectra were computed (Figs. 55 and 57).

The recorded spectrum shows extremely low intensity and a very poor signal-to-noise ratio, making the OH rotational features almost completely indistinguishable from the noise. The most significant factor suppressing the OH emission signal in a sterno flame is the high concentration of collisional quenchers in the burnt gas products. Sterno is typically composed of denatured alcohol (often ethanol or methanol). Alcohol fuels produce large amounts of H₂O (water vapor) and CO₂ as combustion products. Both H₂O and CO₂ are highly effective collisional quenchers. They rapidly collide with the excited OH radical and de-excite it (transferring the energy as heat) before the OH can emit a photon. This fast de-excitation drastically reduces the quantum efficiency of the OH emission process, leading to a much lower overall emission intensity compared to flames with cleaner products.



Figs. 54 & 55: Spectrum intensities measured (left) and emission data (right) at the flame height of 0.25"

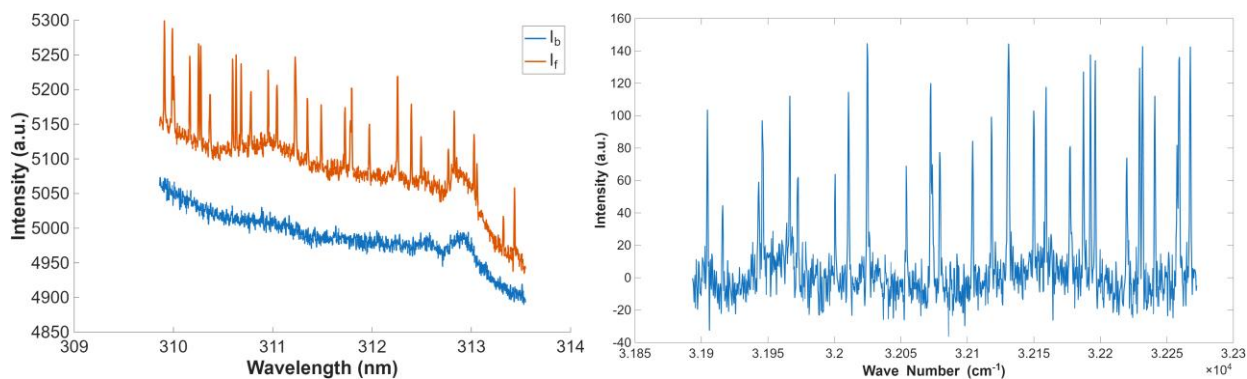


Figs. 56 & 57: Spectrum intensities measured (left) and emission data (right) at the flame height of 0.5''

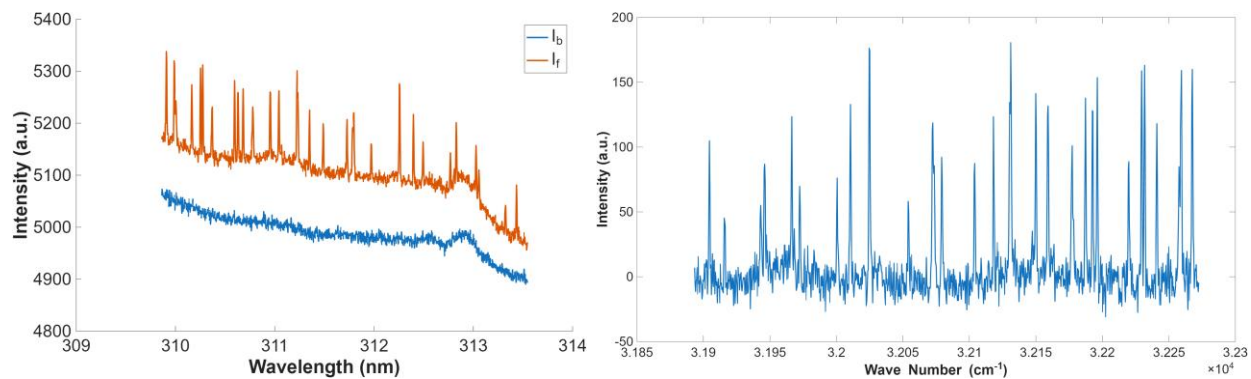
5.12. Emission spectrum analysis of hexamine flame

For the hexamine flame, emission measurements were performed at two vertical locations above the burner. The two required spectra I_f and I_b were acquired as shown in Figs. 58 and 60. Following zero-baseline correction and wavelength calibration, the emission spectra were computed (Figs. 59 and 61), and multiple isolated OH A-X (0-0) rotational lines were integrated to obtain their equivalent widths. These values were then used to construct the Boltzmann plots for each measurement height (Figs. 62 and 63). From the slope of each plot, the rotational temperature was extracted.

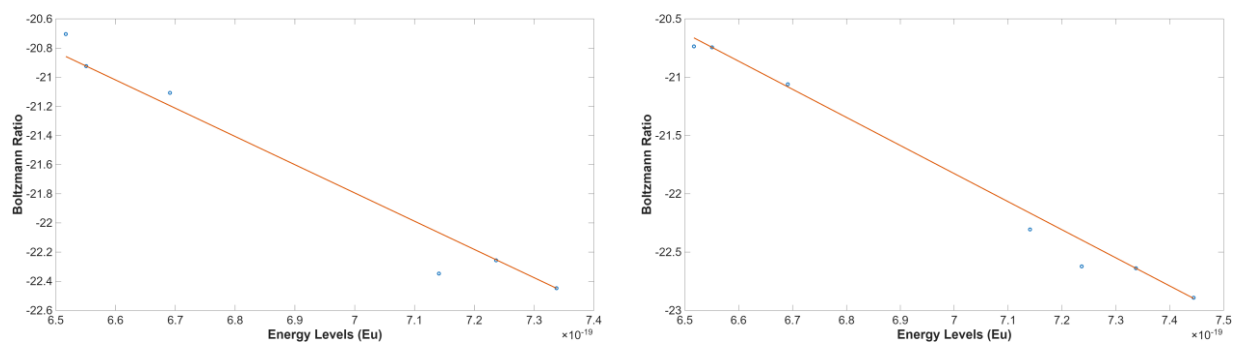
At the lower sampling height of 0.25'' in the butane flame, the Boltzmann fit yielded a temperature of approximately 3739 K, while at the higher measurement position (0.5''), the fitted temperature (3008 K) decreased.



Figs. 58 & 59: Spectrum intensities measured (left) and emission data (right) at the flame height of 0.25''



Figs. 60 & 61: Spectrum intensities measured (left) and emission data (right) at the flame height of 0.5"

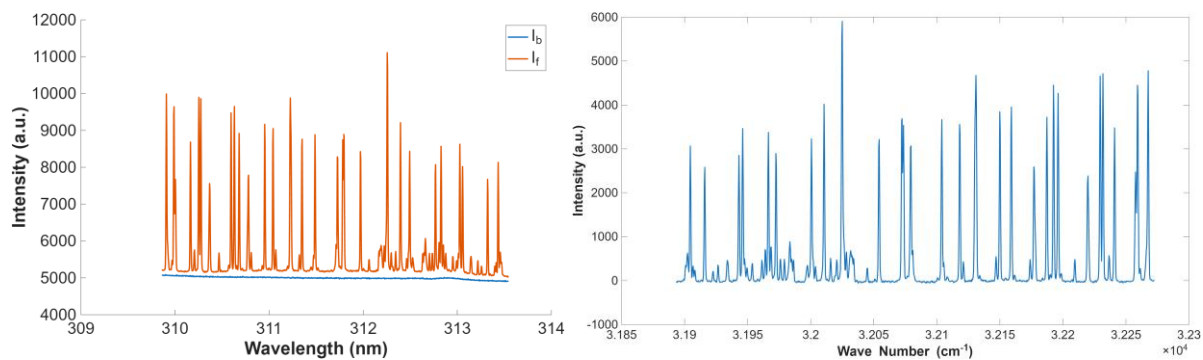


Figs. 62 & 63: Boltzmann plots at flame height of 0.25" (left) and 0.5" (right)

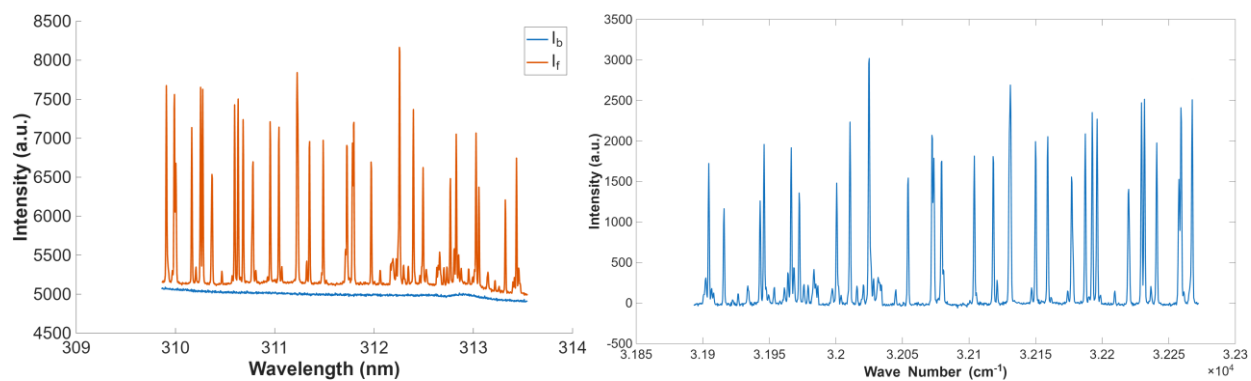
5.13. Emission spectrum analysis of propane flame

For the propane flame, emission measurements were performed at two vertical locations above the burner. The two required spectra I_f and I_b were acquired as shown in Figs. 64 and 66. Following zero-baseline correction and wavelength calibration, the emission spectra were computed (Figs. 65 and 67), and multiple isolated OH A-X (0-0) rotational lines were integrated to obtain their equivalent widths. These values were then used to construct the Boltzmann plots for each measurement height (Figs. 68 and 69). From the slope of each plot, the rotational temperature was extracted.

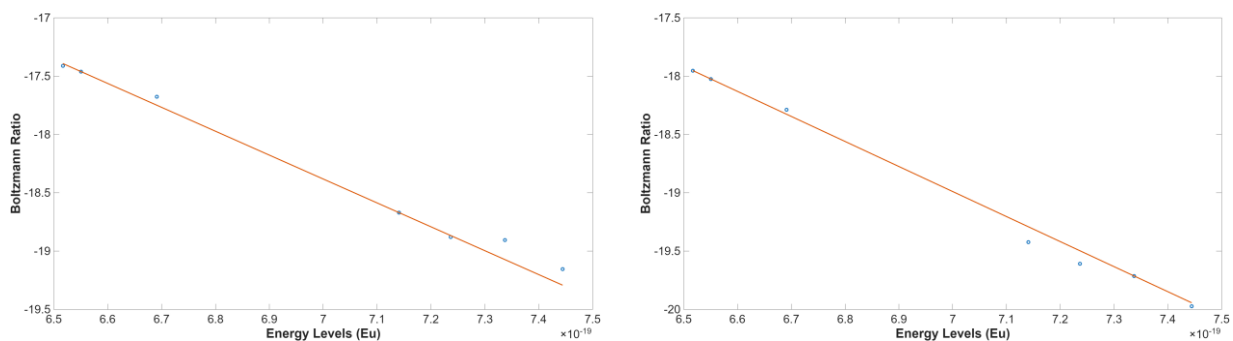
At the lower sampling height of 0.25" in the butane flame, the Boltzmann fit yielded a temperature of approximately 3673 K, while at the higher measurement position (0.5"), the fitted temperature (3378 K) decreased.



Figs. 64 & 65: Spectrum intensities measured (left) and emission data (right) at the flame height of 0.25''



Figs. 66 & 67: Spectrum intensities measured (left) and emission data (right) at the flame height of 0.5''



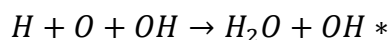
Figs. 68 & 69: Boltzmann plots at flame height of 0.25'' (left) and 0.5'' (right)

5.14. Comparison of absorption vs emission measurements

Across all experiments, the contrast between absorption and emission spectroscopy was clear and instructive. Absorption spectroscopy provided reliable, path-averaged rotational temperatures and OH mole fractions because it probes the ground-state population, which is in Local Thermodynamic Equilibrium (LTE) with the surrounding gas. This population distribution

is minimally affected by soot luminosity or flame brightness. This made it robust for all flame types, including cooler, soot-prone fuels such as candle and sterno, where the OH features are weaker and the background continuum is higher. This proves that absorption excels in providing quantitative species concentration and the true gas kinetic temperature. It serves as the thermal baseline.

Emission spectroscopy, in contrast, is sensitive to the excited-state population and therefore emphasizes hotter, chemically active regions near the flame front. Emission-based temperatures tend to be significantly higher because the excited OH radical is predominantly formed through chemiluminescence [5,6]



This highly exothermic chemical reaction deposits excess energy directly into the rotational mode of the OH molecule, causing the population to deviate wildly from the gas kinetic temperature. The calculated temperature reflects this chemical excitation energy, not the true thermal temperature of the gas. As a result, emission-based temperatures tended to be higher and more spatially localized, with line intensities strongly dependent on local reaction rates and chemiluminescence. Thus, emission is a powerful tool for revealing local flame structure and radiative behavior. It maps the chemical reaction front and the rate of heat release.

Together, the two techniques offered complementary perspectives: absorption excels in quantitative species and temperature retrieval, while emission reveals local flame structure and radiative behavior. Furthermore, attempting to compute the absolute OH mole fraction from emission data was found to be not worthwhile because reliable absorption data were available. This is because the OH emission intensity is non-linearly dependent on the local reaction rate and is severely complicated by collisional quenching. The extreme emission temperature values

confirm that the necessary assumption for quantitative concentration analysis (LTE) is violated. Therefore, absorption provides the robust, direct measure of the total OH concentration, while emission is best reserved for its unique role as a tracer of chemical kinetics.

5.15. *Comparison of the different flame characteristics*

Clear differences among the various flame types were also evident from the spectroscopic measurements. Gaseous fuels such as propane and butane produced strong, well-resolved OH A-X transitions with relatively low continuum background, enabling accurate Boltzmann fitting and consistent OH mole-fraction estimates. These flames showed moderate cooling with height, reflecting their clean, premixed-like reaction zones. In contrast, solid and gel fuels hexamine, candle, and sterno, displayed much weaker OH absorption signals and significantly stronger continuum emission arising from soot (candle), thermal glow (sterno), or fuel decomposition products (hexamine). These flames exhibited lower temperatures overall and a more pronounced reduction in both temperature and OH concentration with height due to their diffusion-controlled, oxygen-limited combustion. The combined observations emphasize that fuel chemistry and flame structure fundamentally influence the spectral signature, with gaseous fuels yielding clean, analyzable spectra and diffusion flames requiring more careful baseline correction but still producing meaningful rotational and concentration trends.

Table 2 summarizes the results of the measurements obtained for the different flames using absorption and emission methods.

		Butane	Candle	Sterno	Hexamine	Propane
0.25"	Temp. Absorption	2020 K	1810 K	1773 K	2075 K	2184 K
	Temp. Emission	3902 K	High background signal from soot	Low signal from high quenching	3739 K	3673 K
	X(OH)	0.0167	0.0056	0.0088	0.0119	0.0430
0.5"	Temp. Absorption	1988 K	1030 K	1746 K	1959 K	2122 K
	Temp. Emission	3865 K	High background signal from soot	Low signal from high quenching	3008 K	3378 K
	X(OH)	0.0162	0.00056	0.0083	0.0102	0.0200

Table 2: OH spectroscopy measurement results

CHAPTER 6: CONCLUSION

This study successfully applied both absorption and emission laser-spectroscopic methods to characterize OH rotational temperature and mole fraction across a range of flames, including gaseous (butane, propane) and diffusion-based solid/gel fuels (candle, sterno, hexamine). Absorption spectroscopy proved to be a robust tool for quantitative diagnostics, yielding reliable Boltzmann temperatures and OH concentrations even in soot-laden or low-radical flames, while emission spectroscopy provided complementary insight into excited-state populations and local radiative behavior. Across all fuels, the measurements revealed consistent physical trends: higher temperatures and OH concentrations near the base of the flame, followed by cooling and radical decay with height. Differences between fuels were also clearly observed. Clean-burning gaseous fuels produced strong, well-resolved OH features, whereas diffusion-controlled solid and gel fuels exhibited weaker OH transitions and stronger continuum backgrounds due to soot or thermal emissions. Overall, the combination of measurement techniques and multi-fuel comparison demonstrated not only the versatility of OH spectroscopy for combustion diagnostics but also the strong influence of fuel chemistry and flame structure on observed spectral behavior. This work provides a reliable framework for using spectroscopic tools to study real-world combustion systems and their underlying reactive-flow physics.

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